



Understanding autism and its treatment: The child's perspective

Yael Karni-Visel^{a,*}, Nehami Baum^a, Mitchell Schertz^{b,c}

^a The Louis and Gabi Weisfeld School of Social Work, Bar Ilan University, Ramat Gan, Israel

^b Child Development and Pediatric Neurology Service, Meuhedet Health Services, Israel

^c Ruth and Bruce Rappaport Faculty of Medicine, Technion – Israel Institute of Technology, Haifa, Israel

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ABSTRACT

Introduction: While parents' and professionals' perceptions regarding children with autism spectrum disorder (ASD) have been studied extensively, limited data regarding the perspectives of children with ASD on their needs and the challenges they face are available. The study aimed to examine how children with ASD understand their condition and the aims of the interventions they undergo.

Methods: Nineteen children and adolescents (ages 5.7–14.2 years) formally diagnosed with ASD, with borderline to high intelligence (range 70–140), and able to converse verbally were interviewed in person at a child development clinic. A qualitative approach was used to capture children's perceptions of their strengths and challenges and their understanding of a novel ASD treatment. The interview included direct and projective open-ended questions on each topic. Interpretive content analysis was used to evaluate the children's answers. Medical data were extracted from medical records. The children's parents completed questionnaires on their children's disability levels, awareness of ASD diagnosis, and sociodemographic details.

Findings: Children spoke of their embodied sensations and feelings and discussed "normality" vs. "disability." They varied in their awareness of their diagnosis/symptoms, and only one boy named his diagnosis and described its consequences in detail. Most children lacked an understanding of the educational and therapeutic aspects of the goals set for them.

Discussion and conclusions: Children with ASD are aware of their unique emotional and behavioral challenges. Nevertheless, they are frequently excluded from the process of patient information provision and lack an understanding of the goals of interventions. Findings suggest the need to explore developmentally and emotionally adaptive ways to involve children with ASD in discussions of their condition and possible interventions.

1. Introduction

Autism spectrum disorder (ASD) is a lifelong neurodevelopmental disability characterized by persistent deficits in social communication and repetitive behaviors (American Psychiatric Association, 2013). Children with ASD have increased health and developmental needs (Hyman et al., 2020). Consequently, from a young age, they often receive long-term, comprehensive, specialized care from various professionals (Hyman et al., 2020; Tachibana et al., 2017). This care aims to improve their functioning (Deb et al., 2020), enhance their caregivers' well-being, and ameliorate their families' adaptive functioning (e.g., Estes et al., 2019).

The principles of patient-centered care and a client-oriented approach (Makoul, 2001; Wolfe, 2001), as well as legal and ethical recommendations (UNICEF, 1989), emphasize children's need and right

to be involved in matters that affect them in medical contexts (Hagger, 2016). The positive impact of open communication on the delivery of health information is evident in acute or life-threatening situations in childhood (Boeriu et al., 2023; Curran et al., 2019; Hallman and Bellury, 2020) and chronic medical conditions (O'Toole et al., 2015). Experts concur that from a clinical perspective, children's involvement results in better engagement in their treatment (King et al., 2014) and better coping with medical conditions and procedures (Löf et al., 2019; Ranmal et al., 2008; Wangmo et al., 2017).

While children with ASD undergo extensive interventions, research on their views regarding the challenges and therapeutic aims of these is scant (Albaum et al., 2023). The few studies conducted focused mainly on the experiences of older children with ASD at school (Healy et al., 2013) or challenges related to gender differences during adolescence (Morgan, 2023; Oswald et al., 2016). Two studies explored the effects of

* Corresponding author.

E-mail addresses: yaelvisel@gmail.com, yael.karni-visel@biu.ac.il (Y. Karni-Visel).

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the disclosure of a diagnosis of autism or Asperger’s syndrome on adolescents (Huws and Jones, 2008; Preece and Jordan, 2010), notably prior to the publication of the DSM-V ASD diagnostic criteria (Chiarotti and Venerosi, 2020) and social shifts in the perception of disabilities, neurodiversity, and autistic identity (Botha and Gillespie-Lynch, 2022; Dwyer, 2022).

Studies have sought to shed light on children’s coping and experiences from an adult-centric perspective, focusing mainly on parents’ experiences (e.g., Crane et al., 2016; Gentles et al., 2020; Makino et al., 2021; Rasmussen et al., 2020; Reed and Osborne, 2012), professionals’ perspectives (Bejarano-Martín et al., 2020), and retrospective examinations of adults with ASD (Crane et al., 2018; Oredipe et al., 2023; Stagg and Belcher, 2019). Asking adults about children’s experiences has several limitations. Parents and professionals may not always understand what children know or comprehend and may have varying perceptions of their subjective feelings (Scott, 2008; Stein et al., 2019). Caregivers cannot access children’s embodied experiences, and children engage in many activities apart from their caregivers. Furthermore, children sometimes choose not to disclose their experiences or actively conceal their insights, viewpoints, and encounters for various reasons, such as protecting their caregivers (Scott, 2008; Spray et al., 2022), especially when parents hesitate to discuss the diagnosis and its consequences (Crane et al., 2019). Retrospective approaches are limited because when questions about past perception are asked, lived experiences and sensations may be misremembered or blended with later insights (e.g., Oredipe et al., 2023; Stagg and Belcher, 2019).

Recognizing the importance of capturing the perspectives of individuals with ASD while they are actively engaged in therapeutic interventions, we aimed to examine children’s understanding of their condition. Specifically, we explored children’s understanding of their needs and challenges and their perspectives on the treatment they were about to undergo. In addition, we aimed to deepen the understanding of children’s awareness of their condition, the meanings they attribute to it, and their coping strategies.

2. Methods

2.1. Study design

A qualitative approach was used to examine children’s understanding and perception of their medical condition and their understanding of upcoming treatments. We used in-depth semi-structured interviews to shed light on interviewees’ complex subjective experiences and perceptions (Fontana and Frey, 2000).

2.2. Inclusion criteria

Children and adolescents aged four to 16 years registered in the HMO ASD registry with a diagnosis of ASD as per the DSM-V criteria (American Psychiatric Association, 2013) were included. The diagnosis was made by a physician certified in pediatric neurology or psychiatry and a developmental psychologist, as the guidelines of the Israeli Ministry of Health require. The children were Hebrew speakers. Parents or legal guardians provided informed consent.

2.3. Procedure

This study was part of a larger clinical trial examining the efficacy of novel training using a gamification approach in children with ASD. The participants were recruited from among children with ASD who were being treated through the HMO in Israel. After receiving written information about the study, parents who agreed to allow their children to participate signed consent forms. Those who met the entrance criteria were enrolled in the study. All the children enrolled in the clinical trial were interviewed. No parent or child refused to participate.

Medical information was obtained from medical records and

included each participant’s level of intellectual functioning based on a formal cognitive assessment (e.g., WISC-IV) and the level of support needed by each participant per the DSM-V criteria. Interviews lasted, on average, 20 min each (range 10–45 min). One interview was held in two sessions due to the child’s difficulties completing it despite being interested in participating. Interviews were conducted by the first author and one of two experienced research assistants, an experienced developmental physiotherapist, and a psychology student.

The study was conducted from May 2021 to July 2023 at the HMO in Israel, which the children had visited previously as part of their routine care. Two interviewers were present at each interview (with one exception), recording or responding to the child’s requests. Before commencing, we asked each child’s permission to record the interview and explained the study’s purpose. Although most of the interviews were conducted without the parents’ presence, all the children were told that the parents could remain during the interview until they felt comfortable. The door to the space where the interview took place was always unlocked and accessible to the parents.

2.4. Participants

Nineteen children and adolescents (age 5.7–14.2 years, $M = 9.45$, $SD = 2.6$), including two sets of siblings, participated in the study. Based on the parents’ reports, five children were entirely unaware of their developmental challenges, six recognized the presence of difficulties but were unaware of their diagnosis, and five were aware of their diagnosis. In one case, the parents were uncertain regarding the child’s awareness of his condition. All children were Hebrew speakers except one, who was an English speaker. See Tables 1 and 2 for background details.

2.5. The interview guide

The first and second authors developed the interview guide in

Table 1
Personal and familial characteristics.

Variable	Mean(SD); Range	N	%
Age at interview	9.45 (2.64); 5.7–12.7		
Age at first detection	3.43 (1.78); 1–7		
Age at diagnosis	5.87 (2.46); 2.6–11		
Level of cognitive function (IQ)	89.32 (18.23); 70–140		
Gender Boys		15	78.9
Girls		4	21.1
Level of function by medical record			
High function		10	52.6
High–intermediate function		1	5.3
Intermediate function		5	26.3
Severe		3	15.8
Education			
Mainstream class		2	10.5
Mainstream with assistance		8	42.1
Special class in mainstream school		6	31.6
Special education settings		3	15.8
ADL according to parents			
Independent in all respects		2	10.5
Independent in most respects		15	78.9
Need assistance in most/all respects		2	10.6
Family status			
Two parents		12	63.2
Divorced		2	10.5
Single parent		5	26.3
SES			
Above average		6	31.6
Average		7	36.8
Below average		6	31.6
Religiosity			
Secular		13	68.4
Traditional		1	5.3
Orthodox		3	15.8
Ultra-Orthodox		2	10.5

Table 2
Child characteristics.

ID	Siblings (ID)	IQ	Communication function level	Gender	Interview age	DX age	Child's awareness, according to parents	Education	ADL by parents	Past treatments	Current treatments
1		94	High	girl	9.63	2.5	aware of difficulties	2	2	OT + EFT	SLT + EFT
2		76	High	boy	6.23	3.5	unaware of difficulties/diagnosis	2	2	OT + PT + SLT + EFT, Other	EFT
3		91	Intermediate	boy	6.96	5	do not know child's level of awareness	4	2	OT + SLT	–
4		94	High	boy	8.11	6	do not know child's level of awareness	3	2	OT + EFT	OT + EFT
5		76	High (minimal symptoms)	boy	12.38	8	aware of difficulties	1	2	OT + SLT + EFT + Other	OT + SLT + EFT + Other
6		92	High	boy	11.78	4.5	aware of difficulties	2	2	OT + PT + SLT + EFT + Other	OT + EFT
7		70	Intermediate	girl	10.5	4	unaware of difficulties/diagnosis	3	2	OT + PT + SLT + EFT, Other	OT + EFT
8		112	High	boy	7.44	5.5	unaware of difficulties/diagnosis	2	2	OT + PT + SLT + EFT, Other	EFT
9		93	High-intermediate	boy	11.57	10	aware of ASD diagnosis	3	2	EFT	EFT + Other
10		70	High	girl	14.16	3	aware of ASD diagnosis	4	2	OT + SL + EFT	OT + SLT + EFT
11	yes (12)	80	Severe	boy	12.68	7	aware of ASD diagnosis	2	2	OT + SLT + EFT	OT + EFT
12	yes (11)	78	Intermediate	boy	5.7	4	do not know the child's level of awareness	4	3	OT + PT + SLT + EFT	OT + SLT + EFT
13		91	Severe	boy	5.98	5	unaware of difficulties/diagnosis	3	2	OT + SLT + EFT	OT + SLT + EFT
14		121	High	girl	10.21	9.5	aware of difficulties	1	2	EFT	EFT
15		140	Intermediate	boy	7.3	7	aware of difficulties	2	2	PT + EFT	PT + EFT
16	yes (17)	70	Severe	boy	10.67	3	aware of ASD diagnosis	3	2	OT + SLT + EFT + Other	OT + SLT + EFT
17	yes (16)	80	Intermediate	boy	8.35	7	aware of ASD diagnosis	3	2	OT + SLT + EFT	OT + SLT + EFT
18		83	High	boy	12.75	11	aware of difficulties	2	2	OT + PT + SLT	EFT
19		86	High	boy	7.07	6	unaware of difficulties/diagnosis	2	2	OT + SLT + EFT	OT + SLT + Other

OT = Occupational Therapy; SLT = Speech-language therapy; EFT = Emotional/psychological therapy; PT = Physical therapy; Other = Other therapy (e.g., social skills training, art therapy, hydrotherapy); ADL = Activities of daily living; Independent in all respects = 1; Independent in most respects = 2; Need assistance in most/all respects = 3; Mainstream class = 1; Mainstream with assistance = 2; Special class in mainstream school; Special education settings.

consultation with three content experts (a developmental psychologist, an occupational therapist, and a physician) and then refined it with the assistance of a mother of a child with autism. The interviews included four stages. The first was a brief rapport-building phase in which we asked neutral questions regarding school and family members. Next, we asked children about their challenges at school and home, why they were invited to the clinic, and what they knew about the upcoming intervention. Then, we explored their knowledge regarding the intervention and the source of the information. Finally, we thanked the children for their efforts and invited them to contact us through their parents to ask questions or add information following the meeting.

We used a guide that employed open-ended questions and prompts to interview children about their understanding and perception of their condition and their comprehension of the planned intervention in the clinical trial. To avoid influencing the process of informing children about their medical condition, we posed no direct questions regarding the diagnosis and related symptoms and provided no medical information. The guide was designed for the current study to overcome several challenges posed by the study population, including children's difficulties in establishing rapport, their tendency to experience anxiety, and limited expressive or receptive language (Rasmussen and Pagsberg, 2019). We took several steps to address the study's challenges. Following Tasker and Cisneroz (Tasker and Cisneroz, 2019), we used open-ended questions with "error-free answers" so children would not feel anxious. Importantly, we used only the children's wording without

mentioning the diagnosis. Follow-up questions were allowed, using the children's own words, e.g., "You said you are sensitive to noises. Tell me more about that." (Lamb et al., 2018). We requested factual information (Why have you been invited here today?) and subjective information (How do you feel about it?). We validated all feelings and encouraged children to express themselves as well as possible. We asked children about their wishes by rephrasing the "magic stick" question: "If you had a magic stick through which you could change something about yourself, what would you do with it?". The direct questions were followed by questions about an imaginary child with a profile similar to theirs (age, gender, grade) with ASD symptoms (e.g., hypersensitivity, social difficulties). In addition, children were asked about the imaginary child's (the projective figure's) feelings and motivation for therapeutic intervention.

2.6. Parents' questionnaires

Parents reported sociodemographic information, details regarding the assessment process (age of first referral for a neurodevelopmental assessment at diagnosis), details regarding previous and current number and type of therapeutic interventions (e.g., physiotherapy, occupational therapy, speech and language therapy, psychology, social work), and educational setting (special education class, regular class with/without an assistant) (See Tables 1 and 2). Parents were also questioned regarding their perceptions of their children's awareness of the ASD

diagnosis using the following question: ‘Do you believe your child is cognizant of the diagnosis provided?’ with the following response options: 1. The child is entirely unaware of their developmental challenges. 2. The child recognizes the presence of difficulties without awareness of the diagnosis. 3. The child is aware of the diagnosis. 4. I am uncertain.

2.7. Ethical consideration

The Helsinki Committee of the Meuhedet HMO Research Institute approved the study. All participants’ parents gave written informed consent for their children to participate in the study, and all participants verbally agreed to participate. The participants’ confidentiality and anonymity were maintained by separating their identifying details from the other findings and removing identifying details from the interviews.

2.8. Data analysis

We adhered to an interpretative phenomenological perspective (Shinebourne, 2011), conducting a thematic content analysis (Braun and Clarke, 2006). Two authors read all the transcripts, performed open coding, marked the separate content units of meaning from the interviews, and shared these materials with the other authors, discussing their conceptualizations and critically assessing them until agreement was reached. The findings and theoretical conceptualizations were presented to the third author, who added his input and approved the final findings, ensuring they accurately reflected the children’s experiences. The coders agreed that saturation had been reached when the theoretical categories repeated themselves and were sufficiently grounded. We followed Glaser and Strauss’s approach, which employs a continuous and data-driven sampling process until all constructs of a phenomenon are explored and exhausted to support an emerging theory (Glaser and Strauss, 2017).

2.9. Trustworthiness

We provide a detailed description of each coping experience by quoting excerpts from the participants’ responses to promote credibility (Creswell and Creswell, 2017). We used investigator triangulation to determine the trustworthiness of the themes (Creswell and Miller, 2000). Regarding reflexivity, the research team included a social worker who specialized in interviewing children, including those with neurodevelopmental disabilities, a neurodevelopmental pediatrician, and a researcher with a clinical background in social work and family therapy. Their discussions increased their reflexivity regarding their own positions and beliefs regarding the phenomenon and decreased the potential for bias in data coding and analysis.

3. Findings

In this cohort, children with ASD demonstrated some degree of awareness of their condition, though not necessarily of their diagnosis. Children were insightful and spoke of everyday challenges, especially at school. In contrast, they exhibited confusion regarding belonging to a social group and expressed ambivalent thoughts and feelings about their differences from their peers. They recognized the complexity involved in the various forms of educational mainstreaming and spoke of their frustration when attempting to fit into mainstream settings.

Interestingly, regardless of age or developmental stage, the vast majority of children showed limited awareness of their diagnosis and a partial understanding of the upcoming therapeutic procedures and their aims. Moreover, they demonstrated minimal knowledge of the research intervention in which they were enrolled. Although some were curious about the program, most were passive and dependent on their parents for information. Children’s perceptions and role expectations are illustrated through three main themes: 1) embodied sensations and feelings, 2) perceptions of “normality” vs. “disability” in educational settings

(regular, special education, mainstream), and 3) comprehension of the therapeutic intervention. See Table 3 for themes, subthemes, and examples.

3.1. Theme I – embodied sensations and feelings

Most of the children shared their likes, dislikes, strengths, and challenges (children #10, #12, and #17 interacted minimally during the interview and commented on other topics, using one- or two-word answers or only responding nonverbally (e.g., gesturing, nodding). Most children referred to struggles in their educational settings, while several mentioned the general challenges of their daily routines. Children varied in their awareness of their unique sensory experiences. Some explained in detail their **(1) sensory sensitivities**, including sensitivity to smells, voices, and tastes, as one child described: “I don’t like meat; I don’t like fish ... Because the smell of meat is (sound indicating disgust).” (I., boy, age seven, #19). Others were aware of being overwhelmed by many sensory stimuli. Moreover, awareness of sensitivity was translated by one interviewee into a specific request to adjust the “here and now” situation of the interview to his needs: “Don’t talk fast because it’s hard for me ... the thread in my shoe is bothering me ... Many things bother me. If I’m hot, it bothers me. If you ask me a lot of questions, it bothers me.” (M., boy, age seven, #4). Some participants reported difficulties experienced as somatic pain: “[In class] ... [the teachers] also ‘drilled’ into my head. My head hurts (S., girl, age 10, #7). **(2)** Most children were aware of their social capabilities and limitations. However, children varied in their insights into their strengths and weaknesses. Some perceived their difficulties as temporary and connected to specific circumstances or individuals, as one boy explained regarding his social situation at the beginning of the school year: “Socially, I was a bit uh ... isolated ... I started sixth grade and I was quite quiet at first, and the children found their own friends and ... [the new friends] were a little tougher [than I]” (Z., boy, age 13, #11). Others acknowledged their limited social skills and wished they had better ones. Although they were aware of their difficulties, their origins were unclear to them, so they could not address them without assistance. Without clear knowledge of the cause of their social difficulties and with

Table 3
Themes, subthemes, and examples.

Themes	Subthemes	Examples
Theme I - Embodied sensations and feelings	(1) Sensory sensitivities	<i>The thread in my shoe is bothering me ... Many things bother me. If I'm hot, it bothers me.</i>
	(2) Social capabilities	<i>It's hard for me to connect ... [although] the interests can sometimes be very similar [to other children's], I don't know why</i>
	(3) Emotional challenges	<i>Sometimes, when I'm not told what's going to happen, I'm so stressed.</i>
Theme II - Awareness and perceptions of normality vs. disability	(1) Defining/naming ASD	<i>It's difficult ... to get used to certain things or [the projective figure] gets used to certain things and can't leave them alone</i>
	(2) Struggling to fit the norm	<i>I didn't go to the last lesson ... it was too difficult for me. Maybe it's not suitable for me, only for ordinary children.</i>
Theme III - Comprehension of the therapeutic intervention	(1) Understanding the framework	<i>[The staff] can help him make friends.</i>
	(2) Practical requirements and considerations	<i>I don't know what they'll do with me here. I think I'll be mind-controlled.</i>
	(3) Engagement in the healthcare dialogue	<i>I didn't even know we were coming here.</i>

a limited understanding of social interaction facilitators, it is unsurprising that when children were asked the “magic stick” question, they stated they would change their social skills: “I guess I would improve my social skills ... it’s hard for me to connect ... [although] the interests can sometimes be very similar (to other children’s), I don’t know why ...” (E., boy, age 11, #9). (3) Only three children expressed awareness of their **emotional challenges**. One girl wished to regulate her intense emotional reactions: “I want [the magic stick] to change the tantrums I have so I’ll be less angry.” (A., girl, age 10, #16). Another boy could see the “big picture” and understood the connection between his increased need for stability and certainty and his emotional state: “Sometimes, when I’m not told what’s going to happen, I’m so stressed.” (M., boy, age 12, #5). (4) Three children openly discussed the challenges they encountered in adjusting to and complying with “rules,” which they considered external demands that prevented them from pursuing their inner desires. For example, one girl explained her desire to remain true to her own way of doing things: “I don’t like to draw at school. I don’t like being told what to draw ... I like to draw what I want” (A., girl, age 10, #16). One boy wanted to focus on his personal goals and areas of interest: “I don’t like ... all these [math] exercises because I know I won’t use them at all [in the future]. I want to be an archaeologist. I don’t need math” (E., age 11, #9).

While the children mentioned above focused on their own “agendas” and preferred others to adapt to their demands and wishes, one described his efforts to adapt to the environment and other people’s expectations. This boy, who lived in an ultra-Orthodox community and routinely participated in a custom that involved wearing a crown when celebrating an educational achievement, shared his insights into his engagement in regulating his anxiety and gradually desensitizing himself to the source of his discomfort: “There is something that bothers me ... crowns [pictures of crowns]. I used to have to throw it away [the photo] because it bothered me a lot, but now I’ve gotten used to it a little ... Sometimes I stand by it ... [and] look at it alone” (M., boy, age 12, #5).

3.2. Theme II - awareness and perceptions of normality vs. disability

3.2.1. Defining ASD

Ten of the 19 children expressed some degree of awareness of their differences. The children’s self-perception can be described on a continuum, from those who used a vague image of their disability to one child who was aware of his diagnosis and its characteristics. Children used various positive, negative, or neutral terms to describe the spectrum of personal characteristics (e.g., different, special, disabled). Regarding the feeling of being different, children found it easier to refer to the emotions of a projective figure than to themselves. A 10-year-old girl with low communication skills could not explain her feelings (her answer was “I don’t know.”) but described the feelings of a projective figure: “She feels **different**, she is very different. Not like everyone else” (S., girl, age 10, #7). Another child described his unique sensitivity more positively: “I have ... **special** sensations” (M., boy, age 7.5, #4). One child used the social concept “disability” when describing the group of children the interventions targeted but could not position himself clearly in this group. Only one child, who had been diagnosed a year before the interview and was aware of his diagnosis, used medical terms (e.g., autism, diagnosis, related symptoms) rather than social jargon: “... Many things [in the story] are related to ... **autism**. ... [I]t’s difficult ... to get used to certain things or [the projective figure] gets used to certain things and can’t leave them alone, and the part with the smells and sounds is also ... related [to autism]” (E., boy, age 11 years old, #9).

3.2.2. Struggling to fit in

Unlike participants from special education settings, mainstreamed participants expressed awareness of the differences between special and regular educational programs and the mismatch between their own atypical characteristics and the requirements of regular education

settings. For example, a first-grader attempted to reconcile this discrepancy (special vs. regular) using an idiosyncratic term: “I’m in a regular school, not special education. [This school is] for ordinary children and also for me. I’m not ordinary, I have a happy brain.” (A., boy, age 7, #2).

In Israel, children with typical development usually study with the same classmates for several years. Mainstreamed children with ASD are required to join specific lessons in different classes in which other children have already formed strong social bonds. Their degree of integration into regular educational settings concerned children who attended school in more than one type of setting and were aware they were expected to fit in with their peers (i.e., gradually increase the amount of time they attended mainstream classes with typically developing children). This expectation, however, appeared to impose an emotional and practical burden on them. One girl who attended mainstream classes explained her difficulties adjusting to and fitting into a class of typical children. She described her struggle to study and explained how it emphasized her sense of not belonging:

Last year, I started the mainstreaming program in regular classes, and ... it was more or less fine. But this year, it has become very difficult ... Two days ago, I had three “mainstream” lessons, and I didn’t go to the last because it was too difficult for me. ... I didn’t understand anything there. Maybe it’s not suitable for me, only for ordinary children. (S., age 10, #7).

3.3. Theme III – comprehension of the therapeutic intervention

Like typically developing children, children with ASD depend on their parents, who control the information regarding their medical situation and treatment and decide what to share with them. The parents usually have the exclusive authority over what information to share. In the interviews, most children demonstrated a limited understanding of many aspects of the upcoming intervention, including its framework (setting), aim, structure (e.g., being interviewed, providing a saliva sample, wearing an EEG helmet, playing with computerized software), and their role in it (e.g., their ability choose to participate, quit, or request information).

3.4. Understanding the framework (therapeutic setting and intervention)

Children had only a vague understanding of the clinic’s purpose and goals, and some had unrealistic expectations. For example, to the question regarding why he thought the projective figure should come to this clinic, E., a seven-year-old boy, answered, “[The staff] can help [him] make friends” (#15). Another girl believed that therapy would provide functional assistance in communication: “[The projective girl] is upset ... The program will help her speak like an ordinary person” (E., girl, age nine and a half, #1).

Most children had limited information about the upcoming therapeutic program. In response to direct or indirect questions, 12 children responded that they did not know why they had come to the clinic or remained silent. Only seven knew that the intervention was related to their difficulties and challenges, as one child explained: “... Mom told me ... I could put something on my head to help me” (E., girl, age nine and a half, #1). One 10-year-old girl had only a vague notion that the therapeutic aim was “not to be shy.” One child who spoke freely about his ASD diagnosis thought that the intervention was part of the routine assessment at the clinic.

3.5. Practical requirements and considerations

Children varied in their knowledge of the tasks planned for them, although they were included in the consent form parents signed and explained in detail by the physician to the parents at the preliminary meeting. Most children knew that the research plan included playing

with computerized software and expressed excitement about this activity. A thirteen-year-old boy knew details about the games and was aware that he would play them at home with his parents. The limited information left room for the children's imaginations and ideas. One boy stated, "I don't know what they'll do with me here. I think I'll be mind-controlled." (E., boy, age seven, #3). Only one child reported accurate information regarding the schedule and content of the meeting. Before the intervention, his mother had given him details (the meeting plan and the physician's name) and her subjective evaluation of the child's chances of benefitting from it. Providing information allowed the child to participate in the process actively and request additional information, as he explained:

My mom thought it was a good idea, this project. Because she thinks I'm a smart kid and it can work. She told me that ... when I come here ... Dr. XX will give me [a] game to play with for ten minutes ... and I need to get better at it, and we'll have a meeting (the interview). I asked how long it would take. (Z., boy, age 12, #11)

3.6. Engagement in the healthcare dialogue

The differences between children's levels of engagement pointed to the parents' role as moderators and gatekeepers of information and decision-making. The children's responses indicated they had received only limited information regarding their upcoming treatment: "I don't know [why we came here]. I didn't even know we were coming here." (A., boy, age seven, #2). Some seemed surprised to learn they might be more active in their treatment and indicated they had not asked why they were attending the clinic. One girl had deliberately not been informed of her ASD diagnosis and was only aware of her ADHD diagnosis. Her mother requested that the staff refrain from discussing the diagnosis with her daughter. While the girl had been informed about the intervention program by her parents, she received an incorrect explanation of its aim. In contrast, another child was encouraged by his parents to decide on the extent of his engagement in the program, as the following excerpt shows: "[My parents] said it was some kind of research and I could leave whenever I wanted to if I didn't like it ... not go ... not answer, and they let me read what was written [in the consent form]" (E., boy, age 11, #9).

During the interviews, children were encouraged to share their views in a supportive atmosphere. One child used the interviewer's question, "Is there anything you would like to ask me or tell me?" to prepare herself for the program by asking, "Actually, how does [the intervention] work? Can you tell me before we start?" (E., girl, age nine and a half, #1).

Overall, the study findings revealed that children with ASD vary in terms of their awareness and understanding of their diagnosis and symptoms. They shared their struggles to fit in socially, especially when encountering children without ASD, and generally lacked insight into the educational and therapeutic goals set for them.

4. Discussion

This study investigated how children with ASD perceive their condition and understand what is required of them in their environments. The study revealed three superordinate themes: children's awareness of their sensory and emotional challenges, their perception of diversity, and their understanding of therapeutic interventions. In what follows, we discuss each theme and then present a model of their interplay. We further address the children's perceptions of their condition, social experiences, and knowledge gaps regarding their diagnosis and treatments.

Overall, the children in this study exhibited awareness of their sensory differences, preferences, and difficulties. They openly discussed their emotional challenges, especially their desire to restrain or control their undesired behaviors, such as tantrums or avoidance of specific

triggers. Emotional regulation challenges among children with ASD are common (Jahromi et al., 2012), yet only limited evidence has been collected regarding their perspectives on this aspect as "insider knowledge" (Gilling, 2012). Importantly, some of the children in the current study translated their self-awareness into practical strategies, such as gradual self-exposure and habituation to triggers in situations of hypersensitivity and anxiety. Moreover, several children translated their self-awareness into efforts to improve their participation in social interactions during the interviews (e.g., by asking to slow the interview's pace or explaining their sensitivities). In contrast, the children seemed less aware of their rigidity and its social consequences, perhaps suggesting that understanding the full impact of rigidity in social contexts may be less discussed or more challenging and thus should be better moderated for children with ASD.

Despite the children's comprehensive self-awareness, they tended to display a limited understanding of information managed by adults. Specifically, their awareness of their diagnosis, educational and therapeutic goals, and practical requirements was generally not extensive. They expressed varying levels of awareness of their condition. Their self-perception ranged along a continuum, from a vague idea of difference to comprehensive knowledge regarding the ASD diagnosis and its characteristics. As a previous study showed, children often used normality as a reference point and described their struggle to conform to societal norms (Humphrey and Lewis, 2008). These findings are consistent with adults' retrospective reports of awareness of being different from other children during their childhoods (e.g., Hurlbutt and Chalmers, 2002). Notably, previous studies directly asked children about "having autism" and the ways they construct their identity as autistic individuals (Huws and Jones, 2008; Mesa and Hamilton, 2022; Morgan, 2023). In the current study, we explored children's spontaneous answers without directing them to talk about the labeling process or the diagnosis itself. This approach allowed them to offer self-definitions, articulate their descriptions, and place themselves along the spectrum of their chosen definitions.

The children in this study struggled to make sense of the reasons behind demands, especially in social interactions with neurotypical populations. Generally, they described their efforts to assimilate into the mainstream as a "battle." Previous studies focusing on mainstream school settings highlighted the social cost of 'camouflaging' as a strategy to avoid stigma or align with adults' desire for children with ASD to behave 'normally' (Bagatell, 2007; Mesa and Hamilton, 2022; Schneid and Raz, 2020). In the current study, children also emphasized difficulties adjusting frequently to changing social and academic demands and reported experiencing frustration and loneliness. The findings emphasize the need to explore the costs associated with low awareness of autistic identity and limited opportunities for group belonging (Stevenson et al., 2016).

Ambiguity was also evident regarding the goals of the therapeutic intervention and its procedures. While children described the comfort of the clinical setting, their expectations varied from a vague desire to "make friends" to a narrow aim such as playing computer games. Children differed in terms of their engagement in goal-setting and decision-making. Some had been deliberately misled about the aims of their treatment, while others had the option to refuse or terminate sessions at their discretion, thanks to fairly comprehensive knowledge about the intervention's purpose. The general pediatric literature recognizes the tendency to exclude children from decision-making or goal-setting in health consultations (Bray et al., 2018; Coyne, 2008; Moore and Kirk, 2010; van Woerden et al., 2023). Nonetheless, interest in children's engagement in goal-setting in medical settings is increasing (Teela et al., 2023). Moreover, there is evidence of the positive impact of the involvement of adolescents with ASD in goal-setting on engagement, motivation, and the efficacy of therapy (King and Chiarello, 2014; Pritchard-Wiart and Phelan, 2018; Shevell et al., 2003). Current findings suggest that this is not the case for children with ASD, who tend to be excluded from decision-making and goal-setting despite their frequent

and long-term interactions with healthcare professionals (Eseigbe et al., 2015; Zuckerman et al., 2013).

Our findings suggest that children who demonstrated awareness of their embodied sensations and were involved (by their parents) in information exchange understood their condition. Moreover, they translated their understanding into active engagement in shaping the therapeutic process (see Fig. 1). In contrast, children with limited knowledge regarding their condition, while they may display increased self-awareness, tend to have vague, limited, or inaccurate knowledge regarding the reasons for their interventions (at school or in therapy). Future studies similar to those that have explored conditions such as diabetes (Koopman et al., 2004) should investigate how children with ASD gain knowledge about their condition and interventions and assess the efficacy of strategies for involving children with ASD in their diagnosis and goal-setting decisions that affect their daily lives (e.g., educational and therapy settings).

The current study's findings show that among children with ASD, there are gaps between their self-awareness and understanding of their environments and their knowledge regarding the diagnosis of the condition. These findings expand our knowledge regarding Huws and Jones's concept of 'absent presence' (2008). While the term was originally used to refer to the retrospective view of adolescents regarding the time before diagnosis, the current finding shows that the absence (of awareness) remains evident years after initial detection and formal diagnosis. Most parents reported not discussing the diagnosis openly with their children, and most of the children were unaware of their diagnosis. Autism is considered an "invisible" condition (Thompson-Hodgetts et al., 2020). Thus, in childhood, disclosure of the diagnosis depends on adults. Parents' decision to inform children of their diagnosis is perceived as complex and challenging, given their desire to avoid negative outcomes for their children and families (See Eaton et al., 2017, Kaushik et al., 2016; Thompson-Hodgetts et al., 2020, for reviews on this issue in mental illness and ASD, respectively.) Physicians and other professionals often avoid open communication with the child regarding his condition before the parents are ready, contrary to the principles of patient-centered care (Makoul, 2001; Wolfe, 2001) and evidence of positive clinical outcomes following the disclosure of ASD (Chaudoir and Quinn, 2010; Darling, 2003; Huws and Jones, 2008, 2015; Preece and Jordan, 2010). The limited dialogue with autistic children on their condition, sometimes under strict constraints, points to the complexity of this issue from clinical, societal, and ethical perspectives. An in-depth investigation of the issue with parents, children, and medical professionals is required to implement the principles of person-centered care for children with disabilities.

4.1. Limitations

A limitation of this study is that the findings were not discussed with the participants. Such discussion would have allowed participants to add to or challenge our interpretations. Many parents either did not inform their children of their ASD diagnosis or were unaware of their children's understanding of their condition. Some parents requested that the term ASD not be used in their child's presence. As mentioned, this avoidance allowed children to choose their own words to describe their condition and the reasons for their treatment. Finally, findings are based on a relatively small sample of children who could converse verbally. Therefore, replicating the study with larger samples and including children who communicate using augmentative and/or alternative communication could provide valuable insights.

5. Conclusion

Children with ASD often exhibit proficiency in expressing their personal experiences, needs, and challenges. Nevertheless, they commonly display a limited grasp of their diagnosis and the precise objectives and practical elements of therapeutic interventions. A deeper

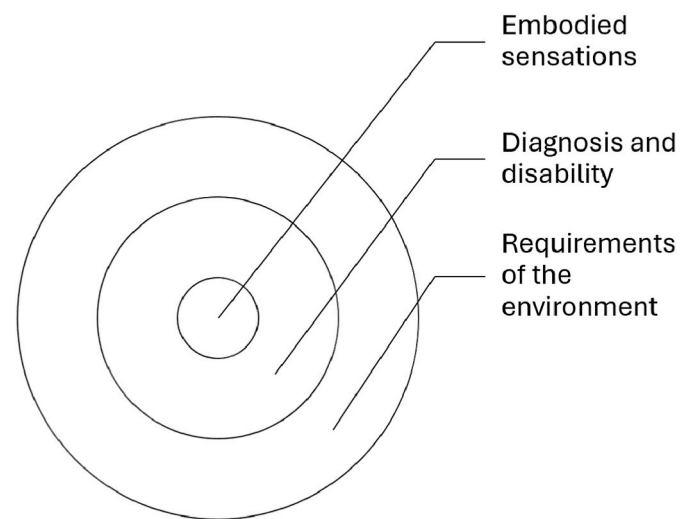


Fig. 1. Children's understanding of ASD at different levels.

understanding of their challenges would significantly enhance their engagement in decision-making and active involvement in therapeutic frameworks. Our findings underscore the critical importance of actively involving children diagnosed with ASD in their care, encouraging their participation in discussing their challenges, and providing tailored information and support to assist them in navigating their environments effectively.

Ethical consideration

The Helsinki Committee of the Meuhedet HMO Research Institute approved the study [01-16-12-19]. All participants' parents gave written informed consent for their children to participate in the study, and all participants verbally agreed to participate. The participants' confidentiality and anonymity were maintained by separating their identifying details from the other findings and removing identifying details from the interviews.

CRediT authorship contribution statement

Yael Karni-Visel: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Nehami Baum:** Writing – review & editing, Methodology, Conceptualization. **Mitchell Schertz:** Writing – review & editing, Project administration.

Data availability

Data will be made available on request.

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